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Unit - 6 Gaseous State

Evaluate yourself:

1. Volume of freon (V_1) = 1.5 dm^3

Pressure (P_1) = 0.3 atm

'T' isconstant

P_2 = 1.2 atm

V_2 = ?

$\therefore P_1V_1 = P_2V_2$

$$\Rightarrow V_2 = \frac{P_1V_1}{P_2}$$

$$= \frac{0.3 \text{ atm} \times 1.5 \text{ dm}^3}{1.2 \text{ atm}}$$

$$= 0.375 \text{ dm}^3$$

$\therefore$ Volume decreased from 1.5 dm^3 to 0.375 dm^3

2. $V_1 = 0.375 \text{ dm}^3$ $V_2 = 0.125$

$P_1 = 1.05$ atm $P_2 = ?$

'T' - Constant

$P_1V_1 = P_2V_2$

$$\therefore P_2 = \frac{P_1V_1}{V_2} = \frac{1.05 \times 0.375}{0.125}$$

$$= 3.15 \text{ atm}$$

3. $V_1 = 3.8 \text{ dm}^3$ $T_2 = 0^0C = 273$ K

$T_1 = ?$ $V_2 = 2.27 \text{ dm}^3$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad T_1 = \left(\frac{T_2}{V_2}\right) \times V_1$$

$$= \frac{273 \text{ K}}{2.27 \text{ dm}^3} \times 3.8 \text{ dm}^3$$

$$T_1 = 457 \text{ K}$$

4. $V_1 = 7.05 \text{ dm}^3$ $V_2 = 2.35 \text{ dm}^3$

$\eta_1 = 0.312$ mol $\eta_2 = ?$

'P' and 'T' are constant

$$\therefore \frac{V_1}{n_1} = \frac{V_2}{n_2}$$

$$\Rightarrow n_2 = \left(\frac{n_1}{V_1}\right) \times V_2$$

$$n_2 = \frac{0.312 \text{ mol}}{7.05 \text{ dm}^3} \times 2.35 \text{ dm}^3$$

$$n_2 = 0.104 \text{ mol}$$

Number of moles exhaled =

0.312 - 0.104 = 0.208 moles

5) $T_1 = 8\ ^0C = 8 + 273 = 281$ K

$P_1 = 6.4$ atm $V_1 = 2.1$ mol

$T_2 = 25^0$ C = 25 + 273 = 298 K

$P_2 = 1$ atm $V_2 = ?$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\Rightarrow V_2 = \left(\frac{P_1V_1}{T_1}\right) \times \frac{T_2}{P_2}$$

$$= \frac{6.4 \text{ atm} \times 2.1 \text{ mol}}{281 \text{ K}} \times \frac{298 \text{ K}}{1 \text{ atm}}$$

$$V_2 = 14.25 \text{ ml}$$

6(a)

$Vo_2 = 12 \text{ dm}^3$ $T = 298$, $P = 1$ atm } Constant

$V_{He} = 46 \text{ dm}^3$

$V_{total} = 5 \text{ dm}^3$

$$P_{O_2} = x_{O_2} \times P_{total}$$

$$x_{O_2} = \frac{n_{O_2}}{n_{O_2} + n_{He}}$$

$$= \frac{0.54}{0.54 + 2.05}$$

$$= \frac{0.54}{2.59} = 0.21$$

$$n_{O_2} = \frac{1 \text{ mol}}{22.4 \text{ L}} \times 12 \text{ L}$$

$$n_{O_2} = 0.54 \text{ mol}$$

$$n_{H_e} = \frac{1 \text{ mol}}{22.4 \text{ L}} \times 46 \text{ L}$$

$$n_{H_e} = 2.05 \text{ mol}$$